HW6

Total Points: 50

1. Let Ω be a bounded domain and $\varphi \in \mathcal{C}(\partial\Omega)$. Let u_φ denote the Perron solution associated with φ, that is,

$$u_\varphi(x) = \sup\{v(x) : v \in \mathcal{C}(\bar{\Omega}),\ v \text{ subharmonic}, \quad v \le \varphi \text{ on } \partial\Omega\}.$$

   (a) [5 points] Use the definition to show that if $\varphi \le \psi$ on $\partial\Omega$, then $u_\varphi \le u_\psi$ in Ω.

   (b) [5 points] Show that

$$|u_\varphi(x) - u_\psi(x)| \le \sup_{\partial\Omega}|\varphi - \psi|, \quad \forall x \in \Omega.$$

[Total for Question 1: 10 points]

2. Let $\Omega \subset \mathbb{R}^d$ be any domain (not necessarily bounded). Let $f \in L^2(\Omega) \cap \mathcal{C}(\Omega)$ and $c > 0$. Define

$$J[u] = \int_\Omega \Big[\frac{1}{2}|\nabla u|^2 + \frac{c}{2}u^2 - fu\Big]\, dx, \quad u \in H^1(\Omega).$$

   (a) [5 points] By computing

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} J[u_* + \varepsilon v],$$

   for any $v \in \mathcal{C}_c^\infty(\Omega)$, show that if $u_* \in \mathcal{C}_0^2(\Omega)$ minimizes $J[u]$ in $H_0^1(\Omega)$, then u_* is a classical solution to

$$-\Delta u + cu = f,\ x \in \Omega, \quad u\big|_{\partial\Omega} = 0.$$

   (b) [5 points] Suppose u_* is a classical solution of the problem above. Show that

$$J[v] - J[u_*] = \frac{1}{2}\int_\Omega |\nabla(v - u_*)|^2 + c|v - u_*|^2\, dx \ge 0.$$

   Conclude that u_* is the unique minimizer of J.

   (c) [10 points] Show that

$$J[v] \ge \frac{1}{2}\|\nabla v\|^2_{L^2(\Omega)} + \frac{c}{4}\|v\|^2_{L^2(\Omega)} - \frac{1}{c}\|f\|^2_{L^2(\Omega)}.$$

   Note that you should not use the Poincaré inequality because Ω is not assumed to be bounded.

[Total for Question 2: 20 points]

3. Let $a, b \in \mathbb{R}$. Consider the variational problem $\inf I[u]$ where

$$I[u] \coloneqq \int_0^1 \sqrt{1 + |u'(x)|^2}\, dx, \quad u \in \mathcal{C}^1[0,1],\ u(0) = a,\ u(1) = b.$$

(a) [5 points] Show that $I[\cdot]$ is convex, that is, $I[\lambda u + (1-\lambda)v] \le \lambda I[u] + (1-\lambda) I[v]$ for $\lambda \in (0,1)$.

(b) [5 points] Compute the first variation of I and solve the Euler–Lagrange equation.

Hint: the solution is a straight line!

[Total for Question 3: 10 points]

4. Let $f \in \mathcal{C}[0,1]$.

(a) [5 points] Show that

$$\lim_{n\to\infty} \int_0^1 f(x) \sin(nx)\, dx = 0.$$

Hint: first prove it for $f \in \mathcal{C}^1[0,1]$ using integration by parts, then use approximation.

(b) [5 points] Let

$$u_n(x) = \begin{cases} 1, & x \in \big(k/n, (k+1/2)/n\big),\ k = 0, 1, \ldots, n-1, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that

$$\lim_{n\to\infty} \int_0^1 f(x) u_n(x)\, dx = \frac{1}{2} \int_0^1 f(x)\, dx.$$

Remark: one can use this to derive $u_n \rightharpoonup \frac{1}{2}$ in $L^2(0,1)$.

[Total for Question 4: 10 points]